

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3445UB0-1



MONDAY, 10 JUNE 2024 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 2: Space, Health and Life
HIGHER TIER

1 hour 30 minutes

ADDITIONAL MATERIALS

- Separate Resource Folder
- Calculator
- Pencil
- Ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	19	
	2.	6	
Section B	3.	5	
	4.	6	
	5.	7	
	6.	10	
	7.	8	
	8.	7	
	9.	7	
	Total	75	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(a) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

You will need to refer to the separate Resource Folder to answer questions 1 and 2.

A Periodic Table is printed on page 24.



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Section A

Answer **all** questions.

Refer to the separate Resource Folder to answer questions **1** and **2**.

1. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct. [2]

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- (b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula. [2]

.....

.....



- (c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

(i) Calculate Malcolm's BMI.

[2]

BMI =

(ii) Calculate Malcolm's maximum heart rate.

[1]

maximum heart rate = bpm

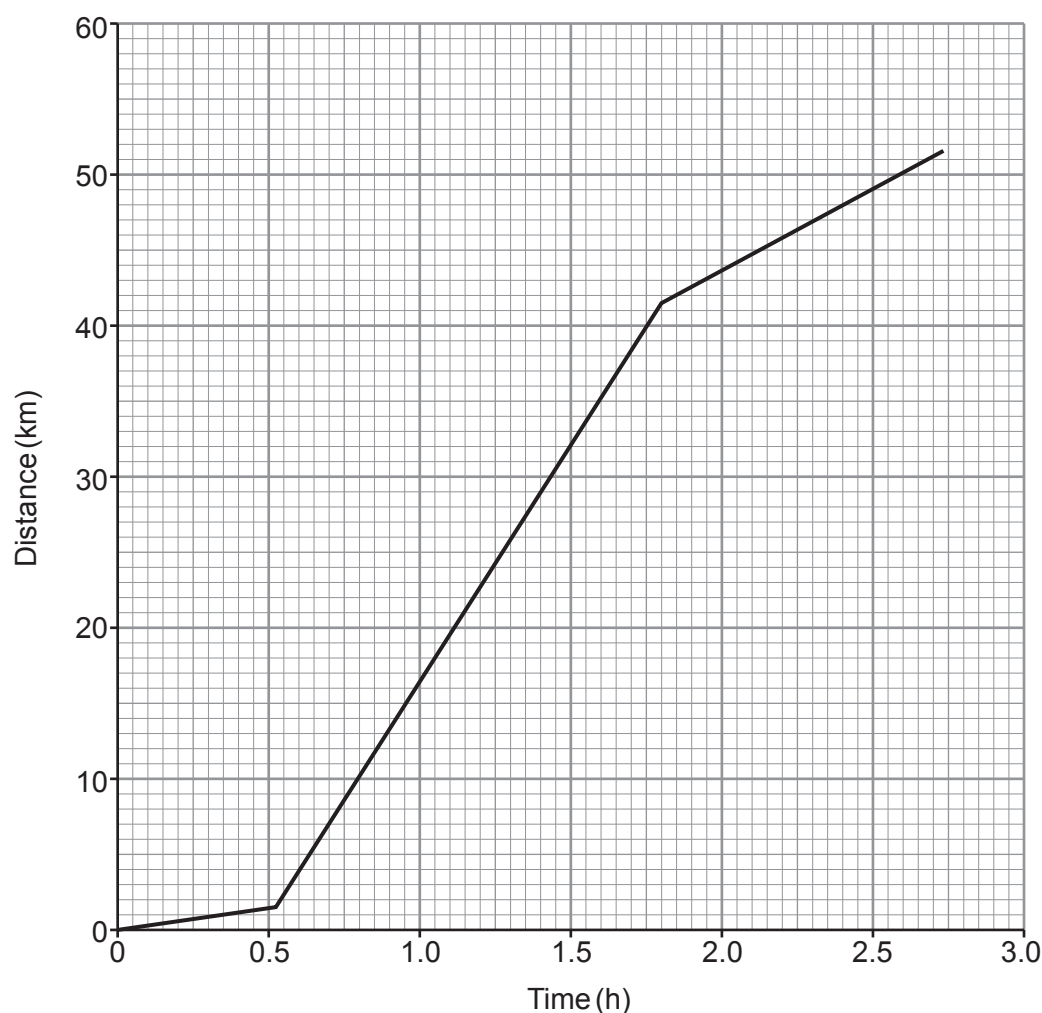
- (d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3 and 4** and on page 4 to determine the range in heart rate she should aim for.

[1]

heart rate range is from bpm to bpm.



- (e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.



Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon:

Reason 1:

.....

Reason 2:

.....



- (f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = km/h

- (g) When a triathlete mounts their bike they are travelling at 1.5 m/s. Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = m/s²

- (h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

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2. Use the information in **Table 2** to answer the following questions.

- (a) Calculate the mean increase in heart rate per minute **during exercise**.

[3]

mean increase in heart rate per minute = bpm

- (b) Calculate the time at which the heart rate returns to its resting value after exercise.
Assume that the heart rate drops at a constant rate after exercise.

[3]

time = minutes



Section BAnswer **all** questions.

3. Living things are classified using a system developed by Carl Linnaeus.

(a) There are 5 kingdoms. These include single-celled organisms and plants.

Name **two** other kingdoms.

[2]

..... and

(b) Linnaeus' system further ranked living things to a species level. This is illustrated below. Complete the sequence. [2]

kingdom → phylum → → order → → genus → species.

(c) *Panthera leo* is the scientific name for a lion.

State the advantage of using scientific names instead of common names.

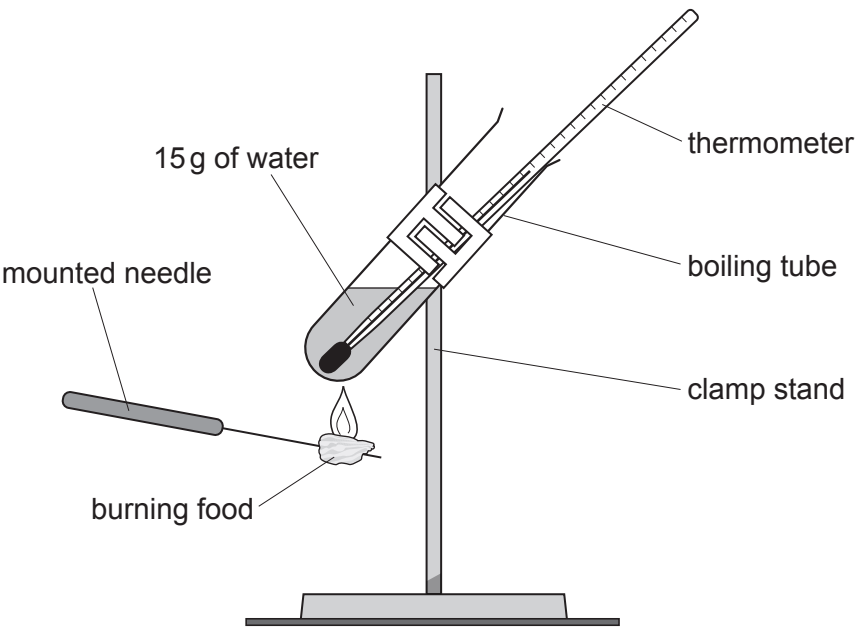
[1]

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4. Students were investigating the energy content of food. They heated 15g of water with each sample of food.



Their results are shown in the table below.

Food	Mass of food burnt (g)	Initial water temperature (°C)	Final water temperature (°C)	Rise in temperature of water (°C)
bread	2.5	18	23	5
cheese	4.0		31	14
cornflakes	3.0	17	29	12
oatie biscuit	5.0	18	25	7

(a) Complete the table. [1]

(b) Use the equation:
energy released (J) = mass of water (g) × 4.2 × temperature increase (°C)
to calculate the energy released by the cheese. [2]

energy released = J



- (c) Use the students' results to calculate the energy provided by a snack consisting of 20 g of cheese and 15 g of oatie biscuits. [3]

energy = J

6

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5. All drugs can have side effects. New drugs undergo testing before they can be prescribed by doctors.

(a) State the meaning of each of the following terms used in drug development.

(i) Blind study [1]

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.....

(ii) Double-blind study [1]

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(iii) Placebo [1]

.....

.....

(b) Aspirin is a common treatment for patients suffering cardiovascular disease. State **one** positive and **one** negative effect of regularly taking aspirin. [2]

Positive:

.....

Negative:

.....



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(c) A painkiller has 196 possible side effects. The most common side effects and their occurrence are listed below.

Side effect	Occurrence (%)
nausea	12.5
headache	19.5
insomnia	20.7
anxiety	11.5

If there are 100 000 people being treated with this painkiller at any one time, calculate how many are likely to suffer from insomnia. [2]

number of people =

7

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6. (a) The spectrum from a star is crossed by dark lines. This is called an absorption spectrum.

(i) Explain how an absorption spectrum is produced.

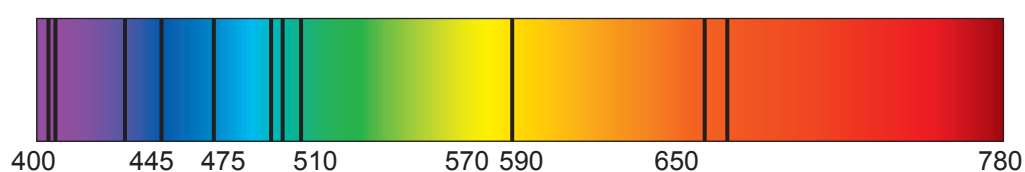
[2]

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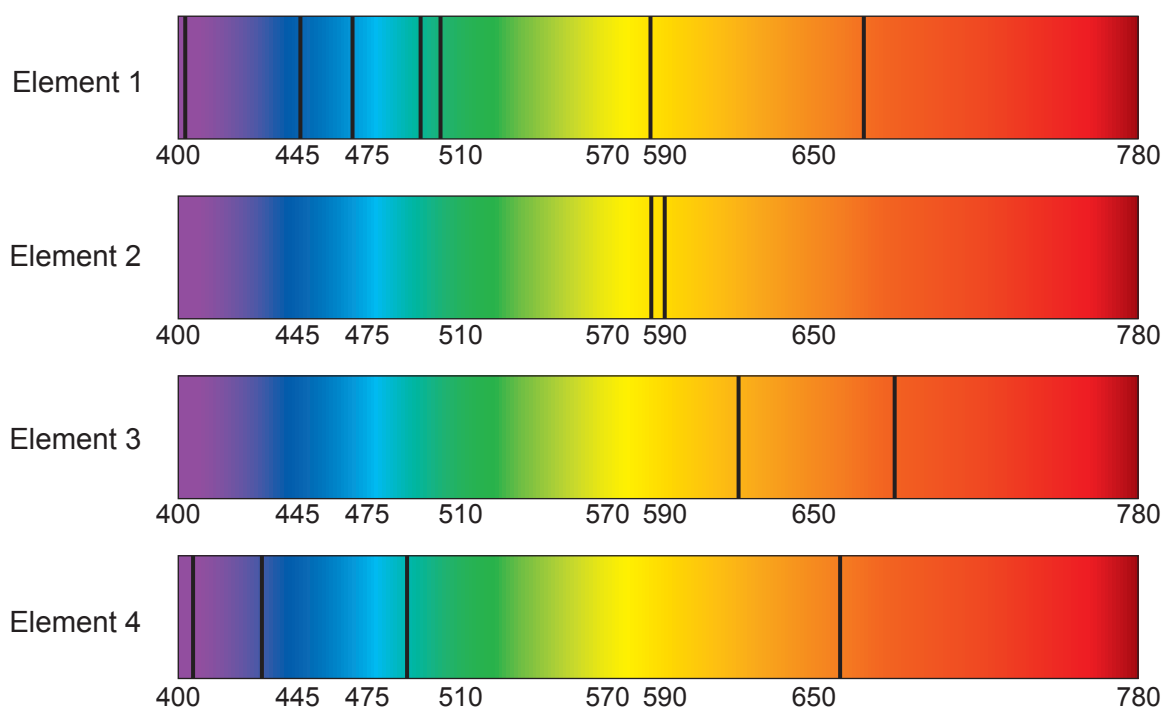
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(ii) An absorption spectrum from a star is shown below.



Here are absorption spectra for different elements.



Complete the table to show whether each element is present in the star.

[2]

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Element	Present in the star (Yes or No)
1	
2	
3	
4	

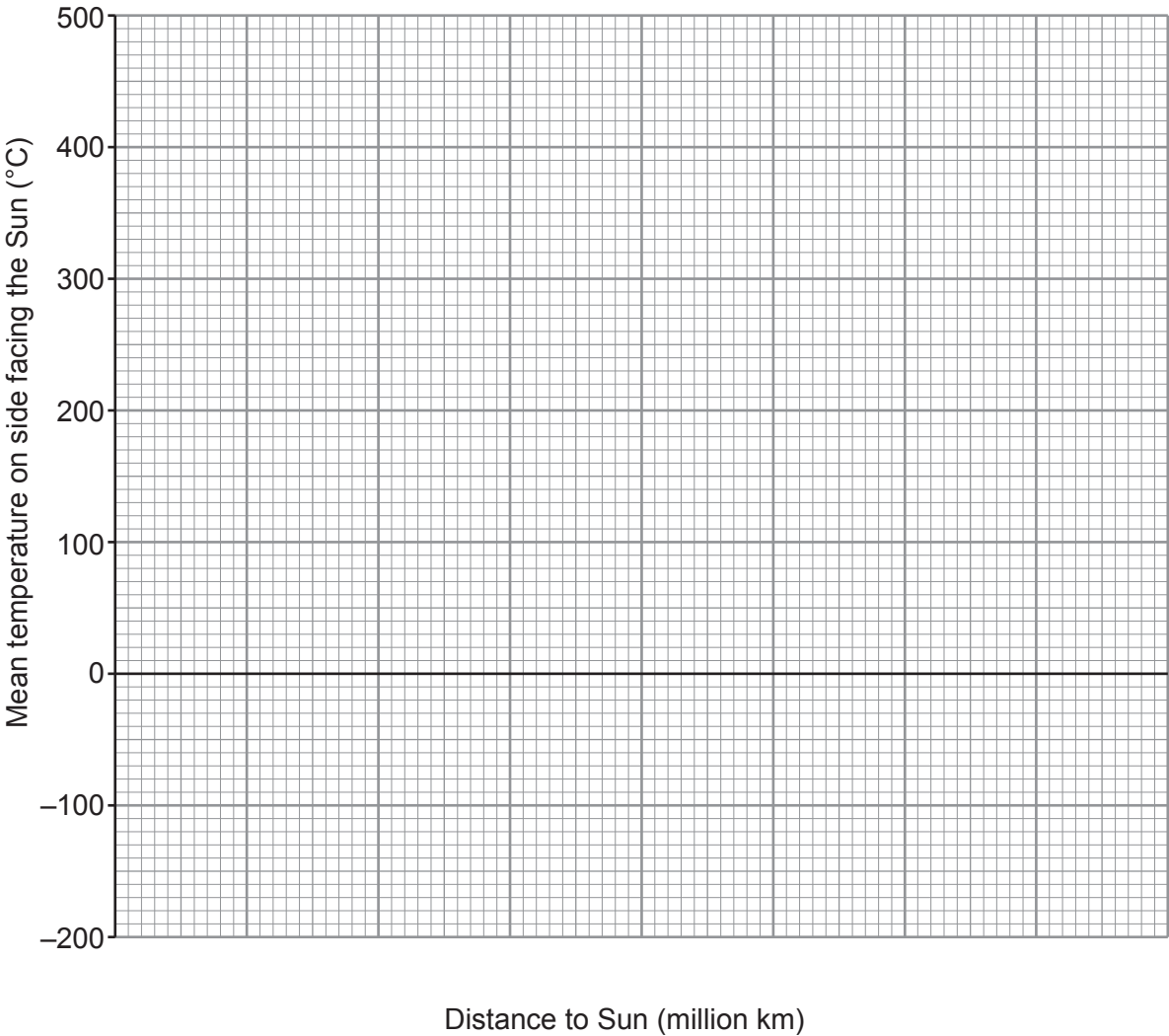


(b) The table gives information about some planets in our solar system.

Planet	Distance to Sun (million km)	Mean temperature on side facing the Sun (°C)
Mercury	60	430
Venus	110	470
Earth	150	20
Mars	230	−20
Jupiter	780	−150

(i) Plot the data on the grid below.

[2]



(ii) Circle the anomalous point on the grid. [1]

(iii) Join the points with a smooth curve. [1]

(iv) Determine the distance from the Sun at which a planet would have a mean temperature to be 0°C . [1]

distance = (million km)

(c) Complete the diagram below to show the shape of the orbit of a comet around the Sun. [1]


Sun



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7. Indicator species may be used in an investigation to monitor water pollution.

(a) Describe how you would investigate water pollution at different points along a stream using indicator species. [6 QER]

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(b) The following chart is available to you.

Pollution level

Indicator species

A Clean water



Stonefly nymph (about 10 mm)



Mayfly nymph (about 20 mm)

B Some pollution



Freshwater shrimp (about 20 mm)



Caddis fly larva (about 10 mm)

C Moderate pollution



Water louse (about 10 mm)



Bloodworm (about 20 mm)

D High pollution



Sludgeworm (about 120 mm)



Rat-tailed maggot (up to 55 mm)

E Very High pollution – no life



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Results from water sampling at two locations are given in the table below.

Species	Total in water sample	
	location 1	location 2
Stonefly nymph	0	0
Caddis fly larva	54	0
Water louse	2	0
Bloodworm	3	0
Sludgeworm	0	80
Rat-tailed maggot	1	65

Use the data in the table to compare the quality of water at the two locations. [2]

8



8. The greenhouse effect is important to stabilise the conditions on Earth needed for life.

(a) Describe how the greenhouse effect occurs.

[3]

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(b) State **two** ways an enhanced greenhouse effect has an impact on Earth.

[2]

1.

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2.

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(c) State **two** ways humans can reduce their dependency on fossil fuels.

[2]

1.

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2.

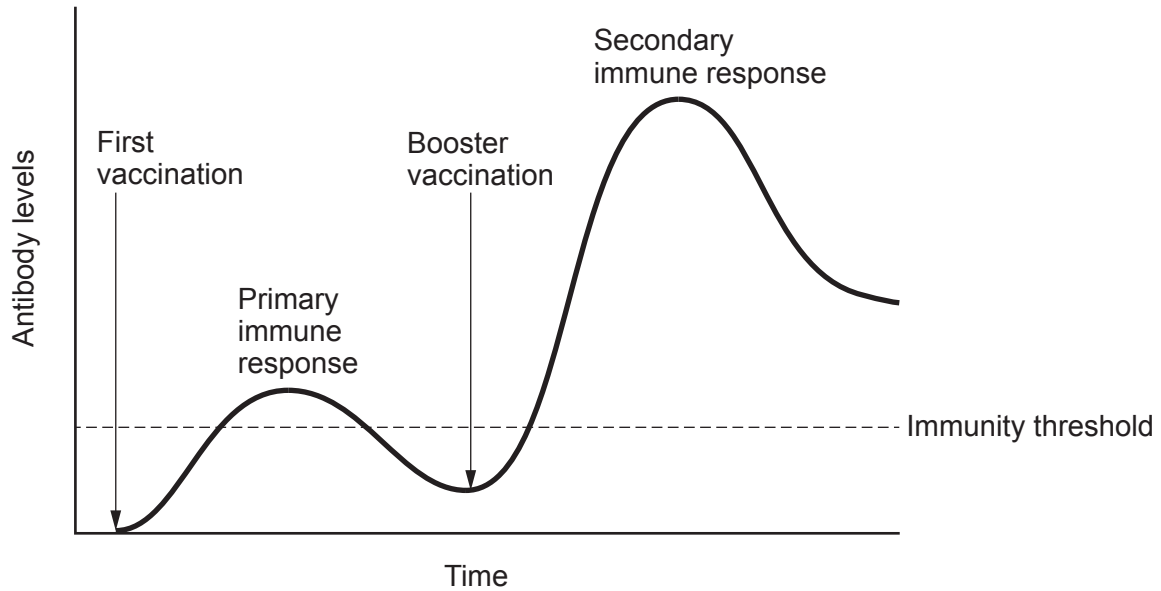
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7



9. Vaccines protect against disease.

- (a) The graph below shows blood antibody levels following a first vaccination and a booster. Once antibody levels rise above a threshold value then there is immunity against the disease.



Compare the immune response following the initial vaccination and the booster vaccination.

[4]

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- (b) Explain how a vaccine protects against specific illnesses.

[3]

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[illegible]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																	
7	9																
Li Lithium 3	Be Beryllium 4																
23	24																
Na Sodium 11	Mg Magnesium 12																
39	40	45	48	51	52	55	56	59	59	63.5	65	70	73	75	79	80	84
K Potassium 19	Ca Calcium 20	Sc Scandium 21	Ti Titanium 22	V Vanadium 23	Cr Chromium 24	Mn Manganese 25	Fe Iron 26	Co Cobalt 27	Ni Nickel 28	Cu Copper 29	Zn Zinc 30	Ga Gallium 31	Ge Germanium 32	As Arsenic 33	Se Selenium 34	Br Bromine 35	Kr Krypton 36
86	88	89	91	93	96	99	101	103	106	108	112	115	119	122	128	127	131
Rb Rubidium 37	Sr Strontium 38	Y Yttrium 39	Zr Zirconium 40	Nb Niobium 41	Mo Molybdenum 42	Tc Technetium 43	Ru Ruthenium 44	Rh Rhodium 45	Pd Palladium 46	Ag Silver 47	Cd Cadmium 48	In Indium 49	Sn Tin 50	Sb Antimony 51	Te Tellurium 52	I Iodine 53	Xe Xenon 54
133	137	139	179	181	184	186	190	192	195	197	201	204	207	209	210	210	222
Cs Caesium 55	Ba Barium 56	La Lanthanum 57	Hf Hafnium 72	Ta Tantalum 73	W Tungsten 74	Re Rhenium 75	Os Osmium 76	Ir Iridium 77	Pt Platinum 78	Au Gold 79	Hg Mercury 80	Tl Thallium 81	Pb Lead 82	Bi Bismuth 83	Po Polonium 84	At Astatine 85	Rn Radon 86
223	226	227															
Fr Francium 87	Ra Radium 88	Ac Actinium 89															

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number